Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “rotation” and “thermal energy” to compare your model or result. Sentence starters are optional.

rotation: turning about an axis

thermal energy: internal energy associated with microscopic motion/interactions

energy budget: an account of all energy stored and transferred

01 Bring back an earlier idea

A ball is thrown vertically upward at 15 m/s from h=0.
Use g=10 and neglect drag.
Find maximum height by energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $mg\Delta h = \Delta K_{translation} + \Delta K_{rotation} + \Delta E_{thermal}$.
Closed energy accounting near Earth; Δh is the positive drop. Determine whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses 6 J of gravitational potential energy but gains only 4 J of translational kinetic energy. The 2 J difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the 2 J remainder explicitly labeled as not yet identified.

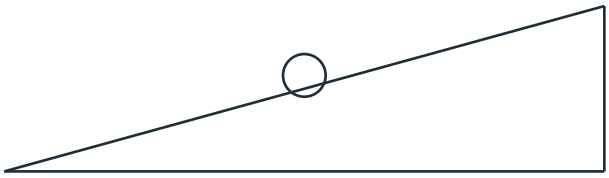
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

A .50 kg cart drops .20 m and reaches 2.5 m/s? **Check** consistency using g=9.8 and starting from rest. What should be checked before calling this energy creation?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Analyze the reasoning

Create a table for repeated trials. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

At fixed momentum magnitude, mass becomes 4 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.

Word help: momentum = mass times velocity, including direction

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Supplied data: m=.50 kg, h=.40 m, v=2.4 m/s, g=9.8. **Find** mgh, K and fraction of initial gravitational energy appearing as translation.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.5 kg cart drops 0.4 m vertically and reaches 2 m/s from rest.
Use g=10.
Find mechanical energy not in translation and average resistive force if all of it is friction over a 1 m path.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3